```
<!DOCTYPE html>
<html lang="en" class="dark">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Integrated Control Center & Operations Hub</title>
<!-- Tailwind CSS -->
<script src="https://cdn.tailwindcss.com"></script>
<script>
tailwind.config = {
darkMode: 'class',
theme: {
extend: {
colors: {
brand: { 50: '#f0fdf4', 500: '#10b981', 600: '#059669', 900: '#064e3b' },
cyber: { bg: '#090d16', panel: '#111827', border:
```

```
'#1f2937', accent: '#38bdf8' }
},
fontFamily: {
sans: ['Inter', 'system-ui', 'sans-serif'],
mono: ['JetBrains Mono', 'Fira Code',
'monospace']
}
}
}
}
</script>
<!-- FontAwesome Icons -->
<link rel="stylesheet" href="https://
cdnjs.cloudflare.com/ajax/libs/font-awesome/
6.4.0/css/all.min.css">
<!-- Google Fonts -->
<link rel="preconnect" href="https://
fonts.googleapis.com">
<link rel="preconnect" href="https://
fonts.gstatic.com" crossorigin>
```

```
<link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800&family=JetBrains+Mono:wght@400;500;700&display=swap" rel="stylesheet">
<style>
body { font-family: 'Inter', sans-serif; background-color: #070b14; color: #f1f5f9; }
.mono-font { font-family: 'JetBrains Mono', monospace; }
.glass-card {
background: rgba(17, 24, 39, 0.8);
backdrop-filter: blur(16px);
-webkit-backdrop-filter: blur(16px);
border: 1px solid rgba(255, 255, 255, 0.08);
}
/* Braille tactile dots */
.braille-dot {
width: 22px;
height: 22px;
border-radius: 50%;
```

```
background-color: #1f2937;
border: 2px solid #4b5563;
transition: all 0.2s ease;
}
.braille-dot.active {
background-color: #34d399;
border-color: #059669;
box-shadow: 0 0 10px rgba(52, 211, 153, 0.7);
}
@keyframes pulse-wave {
0%, 100% { height: 8px; }
50% { height: 36px; }
}
.wave-bar { animation: pulse-wave 1.2s ease-in-out infinite; }
.custom-scrollbar::-webkit-scrollbar { width: 6px; height: 6px; }
.custom-scrollbar::-webkit-scrollbar-track { background: transparent; }
.custom-scrollbar::-webkit-scrollbar-thumb {
```

```
background: rgba(100, 116, 139, 0.4); border-radius: 9999px; }
</style>
</head>
<body class="min-h-screen flex antialiased custom-scrollbar">

<!-- SIDEBAR NAVIGATION -->
<aside class="w-72 bg-slate-900/95 border-r border-slate-800 flex flex-col justify-between hidden md:flex sticky top-0 h-screen z-40">
<div>
<!-- Logo Header -->
<div class="p-6 border-b border-slate-800/80 flex items-center space-x-3">
<div class="w-10 h-10 rounded-xl bg-gradient-to-tr from-cyan-500 via-blue-600 to-indigo-600 flex items-center justify-center text-white shadow-lg shadow-cyan-500/20 font-bold">
<i class="fa-solid fa-terminal"></i>
```

```
</div>
<div>
<h1 class="font-bold text-sm tracking-tight text-white">Operations Hub</h1>
<p class="text-[11px] text-cyan-400 font-mono">Control Center v3.0</p>
</div>
</div>

<!-- Navigation Links -->
<nav class="p-4 space-y-1.5 text-sm font-medium">
<button onclick="switchView('overview')" id="nav-overview" class="sidebar-btn w-full flex items-center space-x-3 px-4 py-3 rounded-xl transition-all bg-cyan-500/10 text-cyan-400 border border-cyan-500/20">
<i class="fa-solid fa-gauge text-base w-5"></i>
<span>Command Overview</span>
</button>
```

```
<button onclick="switchView('fundamentals')"
id="nav-fundamentals" class="sidebar-btn w-full
flex items-center space-x-3 px-4 py-3 rounded-xl
transition-all text-slate-400 hover:text-slate-200
hover:bg-slate-800/60">
<i class="fa-solid fa-microchip text-base w-5"></
i>
<span>Computer Architecture</span>
</button>
<button onclick="switchView('biometric')"
id="nav-biometric" class="sidebar-btn w-full flex
items-center space-x-3 px-4 py-3 rounded-xl
transition-all text-slate-400 hover:text-slate-200
hover:bg-slate-800/60">
<i class="fa-solid fa-shield-halved text-base
w-5"></i>
<span>Biometric Security Lab</span>
</button>
<button onclick="switchView('workstation')"
id="nav-workstation" class="sidebar-btn w-full
```

```
flex items-center space-x-3 px-4 py-3 rounded-xl transition-all text-slate-400 hover:text-slate-200 hover:bg-slate-800/60">
<i class="fa-solid fa-universal-access text-base w-5"></i>
<span>Accessible Workstation</span>
</button>
</nav>
</div>

<!-- Sidebar Footer Status -->
<div class="p-4 border-t border-slate-800/80 m-4 rounded-xl bg-slate-950/60 border border-slate-800/60 text-xs space-y-1.5">
<div class="flex items-center justify-between text-slate-400">
<span>System Status</span>
<span class="flex items-center gap-1.5 text-emerald-400 font-semibold"><span class="w-2 h-2 rounded-full bg-emerald-400 animate-
```

```
pulse"></span> Online</span>
</div>
<div class="text-[10px] text-slate-500 font-
mono">Secure Multi-Modal Session Active</div>
</div>
</aside>

<!-- MAIN CONTENT WRAPPER -->
<div class="flex-1 flex flex-col min-w-0">

<!-- Mobile Header Bar -->
<header class="md:hidden bg-slate-900 border-b
border-slate-800 p-4 flex items-center justify-
between sticky top-0 z-30">
<div class="flex items-center space-x-2">
<i class="fa-solid fa-terminal text-cyan-400"></i>
<span class="font-bold text-sm text-
white">Operations Hub</span>
</div>
<div class="flex space-x-1 overflow-x-auto">
```

```
<button onclick="switchView('overview')"
class="px-2.5 py-1 text-xs rounded bg-
cyan-500/20 text-cyan-400">Home</button>
<button onclick="switchView('fundamentals')"
class="px-2.5 py-1 text-xs rounded bg-slate-800
text-slate-300">Arch</button>
<button onclick="switchView('biometric')"
class="px-2.5 py-1 text-xs rounded bg-slate-800
text-slate-300">Bio</button>
<button onclick="switchView('workstation')"
class="px-2.5 py-1 text-xs rounded bg-slate-800
text-slate-300">Access</button>
</div>
</header>

<!-- VIEW CONTAINER -->
<main class="flex-1 p-4 sm:p-8 max-w-7xl w-full
mx-auto space-y-8">

<!-- ================= VIEW 1: COMMAND
```

```
OVERVIEW ================== -->
<section id="view-overview" class="app-view
space-y-6">
<div class="glass-card rounded-3xl p-6 sm:p-8
relative overflow-hidden">
<div class="absolute -right-20 -top-20 w-80 h-80
bg-cyan-500/10 rounded-full blur-3xl pointer-
events-none"></div>
<div class="relative z-10 max-w-3xl space-y-3">
<span class="px-3 py-1 rounded-full text-xs font-
semibold bg-cyan-950 text-cyan-300 border
border-cyan-800/80">
Unified Operations Center
</span>
<h2 class="text-3xl sm:text-4xl font-extrabold
text-white tracking-tight">
Multi-Domain Intelligence & Simulation Hub
</h2>
<p class="text-slate-300 text-sm sm:text-base
leading-relaxed">
```

Access interactive workstations covering computer science architecture, biometric security stress analysis, and non-visual multi-modal accessibility tools from a single unified interface.
</p>

<!-- Quick Access Cards Grid -->

<div class="w-12 h-12 rounded-xl bg-cyan-500/10 text-cyan-400 flex items-center justify-center text-xl mb-4 group-hover:scale-110

```
transition-transform">
<i class="fa-solid fa-microchip"></i>
</div>
<h3 class="text-lg font-bold text-white
mb-1">Computer Architecture</h3>
<p class="text-xs text-slate-400 leading-relaxed
mb-4">
Examine 4 primary characteristics (Speed,
Accuracy, Diligence, Versatility) and the
evolution across 5 hardware generations.
</p>
</div>
<span class="text-xs font-semibold text-
cyan-400 flex items-center gap-1">Launch
Module <i class="fa-solid fa-arrow-right"></i></
span>
</div>

<div onclick="switchView('biometric')"
class="glass-card rounded-2xl p-6 cursor-pointer
```

```
hover:border-rose-500/50 transition-all group
flex flex-col justify-between">
<div>
<div class="w-12 h-12 rounded-xl bg-
rose-500/10 text-rose-400 flex items-center
justify-center text-xl mb-4 group-hover:scale-110
transition-transform">
<i class="fa-solid fa-shield-halved"></i>
</div>
<h3 class="text-lg font-bold text-white
mb-1">Biometric Security Lab</h3>
<p class="text-xs text-slate-400 leading-relaxed
mb-4">
Test vision vs. voice recognition under
environmental noise, lighting glare, and public
transit stress conditions.
</p>
</div>
<span class="text-xs font-semibold text-
rose-400 flex items-center gap-1">Launch
```

Module </
span>

<div onclick="switchView('workstation')"
class="glass-card rounded-2xl p-6 cursor-pointer
hover:border-emerald-500/50 transition-all group
flex flex-col justify-between">

<div class="w-12 h-12 rounded-xl bg-
emerald-500/10 text-emerald-400 flex items-
center justify-center text-xl mb-4 group-
hover:scale-110 transition-transform">

<h3 class="text-lg font-bold text-white
mb-1">Accessible Workstation</h3>
<p class="text-xs text-slate-400 leading-relaxed
mb-4">
Simulate hands-free voice commands, 6-dot

Perkins Braille chords, spatialized TTS audio, and refreshable tactile displays.
</p>

Launch Module

</section>

<!-- ================== VIEW 2: COMPUTER ARCHITECTURE & FUNDAMENTALS ================== -->
<section id="view-fundamentals" class="app-view space-y-8 hidden">

```
<div>
<h2 class="text-2xl font-bold text-
white">Computer Architecture & Generations</
h2>
<p class="text-xs text-slate-400">Core
Characteristics and Historical Hardware
Evolution</p>
</div>
<button onclick="switchView('overview')"
class="px-3 py-1.5 rounded-lg bg-slate-800
hover:bg-slate-700 text-xs text-slate-300">
← Back to Overview
</button>
</div>

<!-- 4 Characteristics -->
<div class="grid grid-cols-1 md:grid-cols-2 lg:grid-
cols-4 gap-6">
<div class="glass-card rounded-2xl p-5 space-
y-3">
```

 1. Speed

Executes millions to trillions of ops/sec in nanoseconds (10^{-9} s) and picoseconds (10^{-12} s).

Clock:3.5 GHz
<input type="range" min="1" max="50" value="35" class="w-full accent-amber-400 h-1 bg-slate-800" oninput="document.getElementById('arch-speed').innerText = (this.value/10).toFixed(1) + ' GHz'">

```
</div>
<div class="glass-card rounded-2xl p-5 space-y-3">
<div class="text-blue-400 text-lg font-bold"><i class="fa-solid fa-bullseye"></i> 2. Accuracy</div>
<p class="text-xs text-slate-300">100% precision. Governed by GIGO (Garbage In, Garbage Out) principle for flawless execution.</p>
<div class="bg-slate-950 p-2.5 rounded-xl border border-slate-800 text-xs">
<span class="text-[11px] text-slate-400 block mb-1">GIGO Check:</span>
<div class="flex gap-1">
<button onclick="this.nextElementSibling.innerText='Valid: 4'" class="px-2 py-1 bg-emerald-500/20 text-emerald-300 rounded text-[10px]">Clean</button>
```

```
<button
onclick="this.previousElementSibling.previousElementSibling.innerText='Error: NaN'" class="px-2 py-1 bg-rose-500/20 text-rose-300 rounded text-[10px]">Bad</button>
</div>
<div id="arch-gigo" class="text-[11px] text-slate-400 font-mono mt-1">Status: Ready</div>
</div>
</div>
<div class="glass-card rounded-2xl p-5 space-y-3">
<div class="text-emerald-400 text-lg font-bold"><i class="fa-solid fa-arrows-spin"></i> 3. Diligence</div>
<p class="text-xs text-slate-300">Operates continuously 24/7/365 without fatigue, boredom, or loss of concentration.</p>
<div class="bg-slate-950 p-2.5 rounded-xl border border-slate-800 text-xs space-y-1">
```

Tasks:1,420,000

 4. Versatility
Seamless multitasking across vastly different application domains through software programming.

<span class="px-1.5 py-0.5 bg-purple-500/20

```
text-purple-300 rounded text-[10px]">CAD</
span>
<span class="px-1.5 py-0.5 bg-purple-500/20
text-purple-300 rounded text-[10px]">AI / ML</
span>
<span class="px-1.5 py-0.5 bg-purple-500/20
text-purple-300 rounded text-[10px]">Media</
span>
</div>
</div>
</div>

<!-- Generations Selector -->
<div class="glass-card rounded-3xl p-6 sm:p-8
space-y-6">
<div class="flex flex-wrap gap-2">
<button onclick="selectArchGen(1)" id="arch-
btn-1" class="arch-gen-btn px-4 py-2 rounded-xl
text-xs font-bold bg-cyan-600 text-white">1st
Gen (Vacuum Tubes)</button>
```

```
<button onclick="selectArchGen(2)" id="arch-btn-2" class="arch-gen-btn px-4 py-2 rounded-xl text-xs font-bold bg-slate-800 text-slate-300">2nd Gen (Transistors)</button>
<button onclick="selectArchGen(3)" id="arch-btn-3" class="arch-gen-btn px-4 py-2 rounded-xl text-xs font-bold bg-slate-800 text-slate-300">3rd Gen (ICs)</button>
<button onclick="selectArchGen(4)" id="arch-btn-4" class="arch-gen-btn px-4 py-2 rounded-xl text-xs font-bold bg-slate-800 text-slate-300">4th Gen (Microprocessors)</button>
<button onclick="selectArchGen(5)" id="arch-btn-5" class="arch-gen-btn px-4 py-2 rounded-xl text-xs font-bold bg-slate-800 text-slate-300">5th Gen (AI & Quantum)</button>
</div>
<div id="arch-gen-content" class="p-4 bg-slate-950/80 rounded-2xl border border-slate-800 text-sm space-y-3">
```

```
<!-- Populated by JS -->
</div>
</div>
</section>


<!-- ================== VIEW 3: BIOMETRIC SECURITY LAB ================== -->
<section id="view-biometric" class="app-view space-y-8 hidden">
<div class="flex items-center justify-between border-b border-slate-800 pb-4">
<div>
<h2 class="text-2xl font-bold text-white">Biometric Security & Reliability Intelligence</h2>
<p class="text-xs text-slate-400">Vision vs. Voice in Public & High-Noise Environments</p>
</div>
<button onclick="switchView('overview')"
```

```
class="px-3 py-1.5 rounded-lg bg-slate-800
hover:bg-slate-700 text-xs text-slate-300">
← Back to Overview
</button>
</div>

<div class="glass-card rounded-2xl p-6 space-
y-6">
<div class="grid grid-cols-1 md:grid-cols-3
gap-6">
<div class="bg-slate-950 p-4 rounded-xl border
border-slate-800 space-y-2">
<div class="flex justify-between text-xs text-
slate-300"><span>Ambient Noise</span><span
id="b-noise" class="font-mono text-rose-400">65
dB</span></div>
<input type="range" id="b-noise-in" min="30"
max="100" value="65" oninput="updateBioHub()"
class="w-full accent-rose-500">
</div>
```

```
<div class="bg-slate-950 p-4 rounded-xl border
border-slate-800 space-y-2">
<div class="flex justify-between text-xs text-
slate-300"><span>Lighting / Glare</span><span
id="b-light" class="font-mono text-
sky-400">Standard</span></div>
<input type="range" id="b-light-in" min="0"
max="100" value="50" oninput="updateBioHub()"
class="w-full accent-sky-500">
</div>
<div class="bg-slate-950 p-4 rounded-xl border
border-slate-800 space-y-2">
<div class="flex justify-between text-xs text-
slate-300"><span>Crowd Density</span><span
id="b-crowd" class="font-mono text-
indigo-400">Moderate</span></div>
<input type="range" id="b-crowd-in" min="0"
max="100" value="40" oninput="updateBioHub()"
class="w-full accent-indigo-500">
</div>
```

```
</div>

<div class="grid grid-cols-1 md:grid-cols-2
gap-6">
<div class="bg-slate-950/80 p-5 rounded-xl
border border-sky-500/30 space-y-3">
<div class="flex justify-between"><span
class="text-sm font-bold text-sky-400">Vision
(Facial) Success</span><span id="b-v-score"
class="font-mono font-bold text-sky-400">78%</
span></div>
<div class="w-full bg-slate-900 h-2.5 rounded-full
overflow-hidden"><div id="b-v-bar" class="bg-
sky-500 h-full w-[78%]"></div></div>
</div>
<div class="bg-slate-950/80 p-5 rounded-xl
border border-rose-500/30 space-y-3">
<div class="flex justify-between"><span
class="text-sm font-bold text-rose-400">Voice
Recognition Success</span><span id="b-voc-
```

```
score" class="font-mono font-bold text-
rose-400">42%</span></div>
<div class="w-full bg-slate-900 h-2.5 rounded-full
overflow-hidden"><div id="b-voc-bar" class="bg-
rose-500 h-full w-[42%]"></div></div>
</div>
</div>
</div>
</section>


<!-- ================== VIEW 4: ACCESSIBLE
HANDS-FREE WORKSTATION
================== -->
<section id="view-workstation" class="app-view
space-y-8 hidden">
<div class="flex items-center justify-between
border-b border-slate-800 pb-4">
<div>
<h2 class="text-2xl font-bold text-
```

```
white">Accessible Hands-Free Workstation</h2>
<p class="text-xs text-slate-400">Non-Visual
Multi-Modal Integration Simulator</p>
</div>
<button onclick="switchView('overview')"
class="px-3 py-1.5 rounded-lg bg-slate-800
hover:bg-slate-700 text-xs text-slate-300">
← Back to Overview
</button>
</div>

<div class="grid grid-cols-1 lg:grid-cols-2 gap-6">
<!-- Voice Input -->
<div class="glass-card rounded-2xl p-6 space-
y-4">
<div class="flex items-center justify-between
border-b border-slate-800 pb-3">
<h3 class="font-bold text-white flex items-center
gap-2"><i class="
```

```
<!DOCTYPE html>
<html lang="en" class="dark">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-
width, initial-scale=1.0">
<title>Master Integrated Workstation &
Intelligence Hub</title>
<!-- Tailwind CSS -->
<script src="https://cdn.tailwindcss.com"></
script>
<script>
tailwind.config = {
darkMode: 'class',
theme: {
extend: {
colors: {
brand: { 50: '#f0f9ff', 500: '#0284c7', 600:
```

```
'#0369a1', 900: '#0c4a6e' },
tactile: { bg: '#111827', border: '#374151', dot:
'#60a5fa', active: '#34d399' }
},
fontFamily: {
sans: ['Inter', 'system-ui', 'sans-serif'],
mono: ['JetBrains Mono', 'Fira Code',
'monospace']
}
}
}
}
</script>
<!-- FontAwesome Icons -->
<link rel="stylesheet" href="https://
cdnjs.cloudflare.com/ajax/libs/font-awesome/
6.4.0/css/all.min.css">
<!-- Google Fonts -->
<link rel="preconnect" href="https://
fonts.googleapis.com">
```

```
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800&family=JetBrains+Mono:wght@400;500;700&display=swap" rel="stylesheet">
<style>
body { font-family: 'Inter', sans-serif; transition: background-color 0.3s, color 0.3s; }
.mono-font { font-family: 'JetBrains Mono', monospace; }
.glass-panel {
background: rgba(15, 23, 42, 0.75);
backdrop-filter: blur(12px);
-webkit-backdrop-filter: blur(12px);
border: 1px solid rgba(255, 255, 255, 0.1);
}
.light .glass-panel {
background: rgba(255, 255, 255, 0.9);
border: 1px solid rgba(0, 0, 0, 0.08);
```

```
}
/* High Contrast Overrides /
body.high-contrast {
background-color: #000000 !important;
color: #ffff00 !important;
}
body.high-contrast .glass-panel, body.high-
contrast .panel {
background-color: #000000 !important;
border: 3px solid #ffff00 !important;
}
body.high-contrast button {
background-color: #ffff00 !important;
color: #000000 !important;
border: 2px solid #ffffff !important;
font-weight: bold;
}
/ Braille tactile dots */
.braille-dot {
width: 24px;
```

```
height: 24px;
border-radius: 50%;
background-color: #1f2937;
border: 2px solid #4b5563;
transition: all 0.2s ease;
}
.braille-dot.active {
background-color: #34d399;
border-color: #059669;
box-shadow: 0 0 12px rgba(52, 211, 153, 0.6);
}
@keyframes pulse-wave {
0%, 100% { height: 10px; }
50% { height: 45px; }
}
.wave-bar { animation: pulse-wave 1.2s ease-in-out infinite; }
.custom-scrollbar::-webkit-scrollbar { width: 6px; height: 6px; }
.custom-scrollbar::-webkit-scrollbar-track {
```

```
background: transparent; }
.custom-scrollbar::-webkit-scrollbar-thumb {
background: rgba(100, 116, 139, 0.5); border-
radius: 9999px; }
</style>
</head>
<body class="bg-slate-950 text-slate-100 min-h-
screen flex flex-col antialiased selection:bg-
blue-500 selection:text-white custom-scrollbar">

<!-- Master Header -->
<header class="border-b border-slate-800 bg-
slate-900/90 sticky top-0 z-50 backdrop-blur-
md">
<div class="max-w-7xl mx-auto px-4 sm:px-6
lg:px-8 h-20 flex items-center justify-between">
<div class="flex items-center space-x-3">
<div class="w-12 h-12 rounded-2xl bg-gradient-to-
tr from-blue-600 via-indigo-500 to-emerald-400
flex items-center justify-center shadow-lg
```

```
shadow-blue-500/20 text-white text-xl">
<i class="fa-solid fa-microchip"></i>
</div>
<div>
<h1 class="font-bold text-base sm:text-lg
tracking-tight bg-gradient-to-r from-blue-400 via-
teal-300 to-emerald-400 bg-clip-text text-
transparent">
Master Workstation & Intelligence Hub
</h1>
<p class="text-xs text-slate-400">Computer
Science Fundamentals, Biometric Security &
Accessible Workstations</p>
</div>
</div>

<!-- Global Controls -->
<div class="flex items-center space-x-2
sm:space-x-3">
<button onclick="toggleHighContrast()" id="btn-
```

```
hc" class="px-3 py-2 rounded-xl bg-slate-800 hover:bg-slate-700 text-xs font-medium border border-slate-700 flex items-center gap-2 transition-all">
<i class="fa-solid fa-circle-half-stroke text-amber-400"></i> <span class="hidden sm:inline">High Contrast</span>
</button>
<button onclick="toggleTheme()" class="p-2.5 rounded-xl bg-slate-800 hover:bg-slate-700 text-slate-300 border border-slate-700 text-xs" title="Toggle Theme">
<span id="themeIcon">🌙</span>
</button>
</div>
</div>

<!-- Master Tab Navigation Bar -->
<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 flex space-x-2 overflow-x-auto py-2
```

```
border-t border-slate-800/80 no-scrollbar">
<button
onclick="switchMasterTab('fundamentals')"
id="nav-fundamentals" class="master-tab px-4
py-2 rounded-xl text-xs sm:text-sm font-
semibold transition-all bg-blue-600/20 text-
blue-400 border border-blue-500/40 flex items-
center gap-2 whitespace-nowrap shadow-sm">
<i class="fa-solid fa-book-open"></i> Computer
Fundamentals & Architecture
</button>
<button onclick="switchMasterTab('biometric')"
id="nav-biometric" class="master-tab px-4 py-2
rounded-xl text-xs sm:text-sm font-semibold
transition-all text-slate-400 hover:text-slate-200
hover:bg-slate-800 flex items-center gap-2
whitespace-nowrap">
<i class="fa-solid fa-shield-halved"></i>
Biometric Security & Reliability Lab
</button>
```

```
<button
onclick="switchMasterTab('workstation')"
id="nav-workstation" class="master-tab px-4 py-2
rounded-xl text-xs sm:text-sm font-semibold
transition-all text-slate-400 hover:text-slate-200
hover:bg-slate-800 flex items-center gap-2
whitespace-nowrap">
<i class="fa-solid fa-universal-access"></i>
Accessible Hands-Free Workstation
</button>
</div>
</header>

<main class="flex-1 max-w-7xl w-full mx-auto
px-4 sm:px-6 lg:px-8 py-8 space-y-8">

<!--
==========================================
=== -->
<!-- MODULE 1: COMPUTER FUNDAMENTALS &
```

```
ARCHITECTURE -->
<!--
==========
=== -->
<div id="module-fundamentals" class="master-
module space-y-10">
<!-- Hero Banner -->
<section class="relative rounded-3xl overflow-
hidden border border-slate-800 bg-gradient-to-b
from-slate-900 via-slate-900/60 to-slate-950 p-6
sm:p-10 shadow-xl">
<div class="absolute -top-24 -right-24 w-96 h-96
bg-blue-500/10 rounded-full blur-3xl pointer-
events-none"></div>
<div class="relative z-10 max-w-3xl space-y-4">
<span class="inline-flex items-center gap-1.5
px-3 py-1 rounded-full text-xs font-semibold bg-
blue-950 text-blue-300 border border-blue-800">
<i class="fa-solid fa-sparkles text-amber-400"></
i> System Architecture & History Reference
```

<h2 class="text-3xl sm:text-4xl font-extrabold text-white tracking-tight">
Characteristics & Five Generations of Computers
</h2>

Explore the four primary computational pillars (Speed, Accuracy, Diligence, Versatility) and trace the historical evolution from vacuum tubes to AI neural silicon and quantum qubits.

</section>

<!-- 4 Primary Characteristics Grid -->
<section class="space-y-6">
<h3 class="text-xl font-bold text-white flex items-center gap-2">
 4

Primary Characteristics of Modern Computers
</h3>

<!-- 1. Speed -->

<h4 class="text-base font-bold text-white mb-1">1. Speed</h4>
<p class="text-xs text-slate-400 mb-4 leading-relaxed">

Executes millions to trillions of calculations per

second, measured in nanoseconds (10^{-9} s) and picoseconds (10^{-12} s).

</p>

Clock Speed:

3.5 GHz

<input type="range" min="1" max="50" value="35" class="w-full accent-amber-400 h-1.5 bg-slate-800 rounded cursor-pointer" oninput="updateSpeedDemo(this.value)">

<!-- 2. Accuracy -->

<h4 class="text-base font-bold text-white mb-1">2. Accuracy</h4>

100% computational precision. Errors stem exclusively from human programming flaws or bad data (GIGO Principle).

GIGO

```
Simulator:</div>
<div class="flex gap-2">
<button onclick="runGIGODemo(true)"
class="flex-1 py-1 bg-emerald-500/20 text-
emerald-300 rounded text-[11px] font-bold
border border-emerald-500/30">Clean Data</
button>
<button onclick="runGIGODemo(false)"
class="flex-1 py-1 bg-rose-500/20 text-rose-300
rounded text-[11px] font-bold border border-
rose-500/30">Bad Data</button>
</div>
<div id="gigoResult" class="text-[11px] text-
slate-400 font-mono truncate mt-1">Status:
Ready</div>
</div>
</div>
</div>

<!-- 3. Diligence -->
```

<h4 class="text-base font-bold text-white mb-1">3. Diligence</h4>

Operates continuously for days and years without fatigue, boredom, or loss of concentration.

<div class="flex justify-between text-slate-400

```
text-[11px]">
<span>Task Iterations:</span>
<span id="diligenceCounter" class="text-
emerald-400 font-mono font-bold">1,420,000</
span>
</div>
<div class="w-full bg-slate-800 h-2 rounded-full
overflow-hidden">
<div class="bg-emerald-500 h-full w-[85%]"></
div>
</div>
</div>
</div>
</div>

<!-- 4. Versatility -->
<div class="glass-panel rounded-2xl p-6 flex flex-
col justify-between hover:border-blue-500/50
transition-all">
<div>
```

<h4 class="text-base font-bold text-white mb-1">4. Versatility</h4>

Capable of executing vastly different tasks sequentially or simultaneously through software programming.

Multitasking Engine:

CAD</

```
span>
<span class="px-1.5 py-0.5 bg-purple-500/20
text-purple-300 rounded text-[10px]">AI Models</
span>
<span class="px-1.5 py-0.5 bg-purple-500/20
text-purple-300 rounded text-
[10px]">Databases</span>
</div>
</div>
</div>
</div>
</div>
</section>

<!-- 5 Generations of Computers -->
<section class="space-y-6">
<div class="flex flex-col md:flex-row md:items-
center justify-between gap-4">
<h3 class="text-xl font-bold text-white flex items-
center gap-2">
```

```
<i class="fa-solid fa-timeline text-indigo-400"></
i> Five Generations of Computer Hardware
</h3>
<div class="flex flex-wrap gap-1.5 bg-slate-900
p-1 rounded-xl border border-slate-800">
<button onclick="selectGen(1)" id="tabGen1"
class="gen-tab px-3 py-1 rounded-lg text-xs font-
semibold bg-blue-600 text-white">1st Gen</
button>
<button onclick="selectGen(2)" id="tabGen2"
class="gen-tab px-3 py-1 rounded-lg text-xs font-
semibold text-slate-400 hover:text-white">2nd
Gen</button>
<button onclick="selectGen(3)" id="tabGen3"
class="gen-tab px-3 py-1 rounded-lg text-xs font-
semibold text-slate-400 hover:text-white">3rd
Gen</button>
<button onclick="selectGen(4)" id="tabGen4"
class="gen-tab px-3 py-1 rounded-lg text-xs font-
semibold text-slate-400 hover:text-white">4th
```

```
Gen</button>
<button onclick="selectGen(5)" id="tabGen5"
class="gen-tab px-3 py-1 rounded-lg text-xs font-
semibold text-slate-400 hover:text-white">5th
Gen</button>
</div>
</div>

<div id="genDetailPanel" class="glass-panel
rounded-3xl p-6 sm:p-8">
<!-- Populated by JS -->
</div>
</section>

<!-- Interactive Knowledge Quiz -->
<section class="space-y-6">
<div class="glass-panel rounded-3xl p-6 sm:p-8
border border-slate-800 max-w-3xl mx-auto">
<div class="flex justify-between items-center
mb-6 pb-4 border-b border-slate-800">
```

```
<div>
<span class="text-xs font-bold text-blue-400
uppercase">Self Assessment</span>
<h3 class="text-xl font-bold text-
white">Computer Fundamentals Quiz</h3>
</div>
<span id="quizScoreDisplay" class="font-mono
text-emerald-400 font-bold">0 / 4</span>
</div>
<div id="quizContainer"></div>
</div>
</section>
</div>


<!--
==========================================
=== -->
<!-- MODULE 2: BIOMETRIC SECURITY &
RELIABILITY LAB -->
```

```
<!--
==========================================
=== -->
<div id="module-biometric" class="master-
module space-y-8 hidden">
<!-- Hero Banner -->
<section class="relative rounded-3xl overflow-
hidden border border-slate-800 bg-gradient-to-b
from-slate-900 via-slate-900/60 to-slate-950 p-6
sm:p-10 shadow-xl">
<div class="absolute -top-24 -right-24 w-96 h-96
bg-rose-500/10 rounded-full blur-3xl pointer-
events-none"></div>
<div class="relative z-10 max-w-3xl space-y-4">
<span class="inline-flex items-center gap-1.5
px-3 py-1 rounded-full text-xs font-semibold bg-
rose-950 text-rose-300 border border-rose-800">
<i class="fa-solid fa-shield-halved text-
rose-400"></i> Public Environment Security
Analysis
```

<h2 class="text-3xl sm:text-4xl font-extrabold text-white tracking-tight">
Vision vs. Voice Biometric Intelligence & Stress Lab
</h2>

Evaluate real-world operational reliability, environmental noise interference, presentation attack vectors, and multimodal security architectures.

</section>

<!-- Stress Test Simulator -->
<section class="glass-panel rounded-2xl p-6 space-y-6">
<div class="flex flex-col md:flex-row md:items-

```
center justify-between gap-4">
<div>
<h3 class="text-lg font-bold text-
white">Environmental Stress Simulator</h3>
<p class="text-xs text-slate-400">Adjust noise,
lighting, and crowd congestion to test biometric
performance.</p>
</div>
<button onclick="resetBioSim()" class="px-3
py-1.5 rounded-lg bg-slate-800 hover:bg-
slate-700 text-xs text-slate-300 border border-
slate-700">
<i class="fa-solid fa-rotate-right mr-1"></i> Reset
</button>
</div>

<div class="grid grid-cols-1 md:grid-cols-3
gap-6">
<div class="bg-slate-950 p-4 rounded-xl border
border-slate-800 space-y-2">
```

```
<div class="flex justify-between text-xs text-slate-300">
<span>Ambient Noise</span>
<span id="bio-noise-val" class="font-mono text-rose-400">65 dB</span>
</div>
<input type="range" id="bio-noise" min="30" max="100" value="65" oninput="updateBioSimulation()" class="w-full accent-rose-500">
</div>
<div class="bg-slate-950 p-4 rounded-xl border border-slate-800 space-y-2">
<div class="flex justify-between text-xs text-slate-300">
<span>Lighting / Glare</span>
<span id="bio-light-val" class="font-mono text-sky-400">Balanced</span>
</div>
<input type="range" id="bio-light" min="0"
```

```
max="100" value="50"
oninput="updateBioSimulation()" class="w-full
accent-sky-500">
</div>
<div class="bg-slate-950 p-4 rounded-xl border
border-slate-800 space-y-2">
<div class="flex justify-between text-xs text-
slate-300">
<span>Crowd Density</span>
<span id="bio-crowd-val" class="font-mono text-
indigo-400">Moderate</span>
</div>
<input type="range" id="bio-crowd" min="0"
max="100" value="40"
oninput="updateBioSimulation()" class="w-full
accent-indigo-500">
</div>
</div>

<!-- Simulation Results -->
```

```
<div class="grid grid-cols-1 md:grid-cols-2 gap-6 pt-2">
<div class="bg-slate-950/80 p-5 rounded-xl border border-sky-500/30 space-y-3">
<div class="flex justify-between items-center">
<span class="text-sm font-bold text-sky-400"><i class="fa-solid fa-camera mr-2"></i> Vision (Facial) Success</span>
<span id="bio-vision-score" class="font-mono font-bold text-sky-400 text-base">78%</span>
</div>
<div class="w-full bg-slate-900 h-2.5 rounded-full overflow-hidden border border-slate-800">
<div id="bio-vision-bar" class="bg-sky-500 h-full transition-all duration-300" style="width: 78%"></div>
</div>
</div>

<div class="bg-slate-950/80 p-5 rounded-xl
```

```
border border-rose-500/30 space-y-3">
<div class="flex justify-between items-center">
<span class="text-sm font-bold text-rose-400"><i
class="fa-solid fa-microphone mr-2"></i> Voice
Recognition Success</span>
<span id="bio-voice-score" class="font-mono
font-bold text-rose-400 text-base">42%</span>
</div>
<div class="w-full bg-slate-900 h-2.5 rounded-full
overflow-hidden border border-slate-800">
<div id="bio-voice-bar" class="bg-rose-500 h-full
transition-all duration-300" style="width: 42%"></
div>
</div>
</div>
</div>
</section>

<!-- Attack Vectors & Decision Calculator -->
<div class="grid grid-cols-1 lg:grid-cols-2 gap-6">
```

```
<!-- Attack Vectors -->
<div class="glass-panel rounded-2xl p-6 space-y-4">
<h3 class="text-lg font-bold text-white flex items-center gap-2">
<i class="fa-solid fa-skull text-rose-400"></i>
Spoofing & Attack Surfaces
</h3>
<div class="space-y-3 text-xs">
<div class="bg-slate-950 p-3.5 rounded-xl border border-slate-800">
<div class="font-bold text-sky-300 mb-1">3D Silicone Masks (Vision Attack)</div>
<p class="text-slate-400">Bypasses basic 2D RGB cameras unless active infrared (NIR) depth liveness sensing is enforced.</p>
</div>
<div class="bg-slate-950 p-3.5 rounded-xl border border-slate-800">
<div class="font-bold text-rose-300 mb-1">AI
```

```
Voice Cloning / Replay (Voice Attack)</div>
<p class="text-slate-400">Generative neural TTS models synthesize conversational voice replicas from seconds of public audio.</p>
</div>
</div>
</div>

<!-- Decision Matrix Calculator -->
<div class="glass-panel rounded-2xl p-6 space-y-4">
<h3 class="text-lg font-bold text-white flex items-center gap-2">
<i class="fa-solid fa-calculator text-emerald-400"></i> Deployment Decision Matrix
</h3>
<div class="space-y-3 text-xs">
<div>
<label class="text-slate-400 block mb-1">Environment Type:</label>
```

```
<select id="calc-env" class="w-full bg-slate-950 border border-slate-800 rounded-xl p-2.5 text-slate-200">
<option value="outdoor">Outdoor Public Transit Hub (High Noise)</option>
<option value="indoor">Controlled Indoor Access Kiosk</option>
</select>
</div>
<button onclick="calculateDeployment()" class="w-full py-2.5 bg-emerald-600 hover:bg-emerald-500 text-white font-bold rounded-xl shadow transition-all">
Compute Optimal Architecture
</button>
<div id="calc-output" class="p-3 bg-slate-950 rounded-xl border border-emerald-500/30 text-emerald-300 hidden">
<strong>Recommendation:</strong> Vision with 3D IR depth liveness + Contactless token
```

fallback. Avoid pure voice biometrics due to ambient SNR failure.

<!--
==========
=== -->
<!-- MODULE 3: ACCESSIBLE HANDS-FREE WORKSTATION -->
<!--
==========
=== -->

<!-- Hero Banner -->

```
<section class="relative rounded-3xl overflow-hidden border border-slate-800 bg-gradient-to-b from-slate-900 via-slate-900/60 to-slate-950 p-6 sm:p-10 shadow-xl">
<div class="absolute -top-24 -right-24 w-96 h-96 bg-emerald-500/10 rounded-full blur-3xl pointer-events-none"></div>
<div class="relative z-10 max-w-3xl space-y-4">
<span class="inline-flex items-center gap-1.5 px-3 py-1 rounded-full text-xs font-semibold bg-emerald-950 text-emerald-300 border border-emerald-800">
<i class="fa-solid fa-universal-access text-emerald-400"></i> Non-Visual Multi-Modal Integration
</span>
<h2 class="text-3xl sm:text-4xl font-extrabold text-white tracking-tight">
Accessible Hands-Free Workstation Simulator
</h2>
```

```
<p class="text-sm sm:text-base text-slate-300
leading-relaxed">
Test non-visual accessibility modalities
including voice input command parsing, 6-dot
Perkins Braille chord simulation, spatialized
audio routing, and refreshable tactile displays.
</p>
</div>
</section>

<!-- 4 Workstation Modules Grid -->
<div class="grid grid-cols-1 lg:grid-cols-2 gap-6">

<!-- 1. Voice Input Simulator -->
<section class="glass-panel rounded-2xl p-6 flex
flex-col justify-between">
<div>
<div class="flex items-center justify-between
mb-4 pb-3 border-b border-slate-800">
<div class="flex items-center gap-3">
```

```
<div class="w-10 h-10 rounded-xl bg-blue-500/10 border border-blue-500/20 flex items-center justify-center text-blue-400">
<i class="fa-solid fa-microphone-lines text-lg"></i>
</div>
<div>
<h3 class="font-bold text-slate-100 text-base">1. Voice Input Simulator</h3>
<p class="text-xs text-slate-400">Neural STT & Command Parsing</p>
</div>
</div>
<span id="mic-badge" class="px-2.5 py-1 rounded-full text-xs font-semibold bg-slate-800 text-slate-400 border border-slate-700">Standby</span>
</div>

<div class="space-y-4">
```

```
<div class="flex gap-2">
<button onclick="startVoiceRecognition()"
id="btn-listen" class="flex-1 py-3 bg-blue-600
hover:bg-blue-500 text-white font-semibold
rounded-xl text-sm transition-all flex items-
center justify-center gap-2 shadow">
<i class="fa-solid fa-microphone"></i> Start
Voice Command
</button>
<button onclick="simulateVoiceCommand('Read
unread emails')" class="px-3 py-3 bg-slate-800
hover:bg-slate-700 text-slate-300 rounded-xl text-
xs font-medium border border-slate-700">
Preset: "Read Emails"
</button>
</div>

<div class="bg-slate-950 border border-slate-800
rounded-xl p-4">
<div class="text-[11px] text-slate-500 uppercase
```

```
tracking-wider font-semibold mb-1">Live Audio Transcript:</div>
<div id="voice-transcript" class="text-sm font-mono text-blue-300 italic">"Awaiting vocal utterance..."</div>
</div>

<div>
<div class="text-xs text-slate-400 mb-1 font-medium">Command Log:</div>
<div id="command-log" class="bg-slate-950 border border-slate-800 rounded-xl p-3 h-24 overflow-y-auto mono-font text-xs space-y-1 text-slate-300">
<div class="text-slate-600">[System] Voice engine initialized.</div>
</div>
</div>
</div>
</div>
```

```
</section>

<!-- 2. Tactile / Braille Keyboard -->
<section class="glass-panel rounded-2xl p-6 flex flex-col justify-between">
<div>
<div class="flex items-center justify-between mb-4 pb-3 border-b border-slate-800">
<div class="flex items-center gap-3">
<div class="w-10 h-10 rounded-xl bg-teal-500/10 border border-teal-500/20 flex items-center justify-center text-teal-400">
<i class="fa-solid fa-keyboard text-lg"></i>
</div>
<div>
<h3 class="font-bold text-slate-100 text-base">2. Tactile / Braille Keyboard</h3>
<p class="text-xs text-slate-400">6-Dot Perkins Braille Chord Simulator</p>
</div>
```

```
</div>
<span id="chord-display" class="px-3 py-1
rounded-full text-xs font-mono bg-teal-500/10
text-teal-300 border border-teal-500/20 font-
bold">Chord: 000000</span>
</div>

<div class="space-y-4">
<div class="bg-slate-950 border border-slate-800
rounded-xl p-4 flex flex-col items-center justify-
center gap-3">
<div class="grid grid-cols-3 gap-6 my-2">
<div class="flex flex-col items-center
gap-1"><button onclick="toggleBrailleDot(1)"
id="dot-1" class="braille-dot text-xs font-bold text-
slate-400">1</button><span class="text-[10px]
text-slate-500">Dot 1</span></div>
<div class="flex flex-col items-center
gap-1"><button onclick="toggleBrailleDot(4)"
id="dot-4" class="braille-dot text-xs font-bold text-
```

slate-400">4</button><span class="text-[10px]
text-slate-500">Dot 4
<div class="flex flex-col items-center
gap-1"><button onclick="toggleBrailleDot(2)"
id="dot-2" class="braille-dot text-xs font-bold text-
slate-400">2</button><span class="text-[10px]
text-slate-500">Dot 2
<div class="flex flex-col items-center
gap-1"><button onclick="toggleBrailleDot(5)"
id="dot-5" class="braille-dot text-xs font-bold text-
slate-400">5</button><span class="text-[10px]
text-slate-500">Dot 5
<div class="flex flex-col items-center
gap-1"><button onclick="toggleBrailleDot(3)"
id="dot-3" class="braille-dot text-xs font-bold text-
slate-400">3</button><span class="text-[10px]
text-slate-500">Dot 3
<div class="flex flex-col items-center
gap-1"><button onclick="toggleBrailleDot(6)"
id="dot-6" class="braille-dot text-xs font-bold text-

```
slate-400">6</button><span class="text-[10px]
text-slate-500">Dot 6</span></div>
</div>
<div class="flex gap-2 w-full">
<button onclick="submitBrailleChord()"
class="flex-1 py-2 bg-teal-600 hover:bg-teal-500
text-white font-semibold rounded-xl text-xs
shadow">Submit Chord</button>
<button onclick="clearBrailleChord()" class="px-4
py-2 bg-slate-800 hover:bg-slate-700 text-
slate-300 rounded-xl text-xs border border-
slate-700">Clear</button>
</div>
</div>

<div class="bg-slate-950 border border-slate-800
rounded-xl p-3">
<div class="text-[11px] text-slate-500 uppercase
tracking-wider font-semibold mb-1">Tactile
Buffer Output:</div>
```

```
<div id="tactile-output-buffer" class="text-sm
font-mono text-teal-300 min-h-[24px]"></div>
</div>
</div>
</div>
</section>

<!-- 3. Spatialized TTS Output -->
<section class="glass-panel rounded-2xl p-6 flex
flex-col justify-between">
<div>
<div class="flex items-center justify-between
mb-4 pb-3 border-b border-slate-800">
<div class="flex items-center gap-3">
<div class="w-10 h-10 rounded-xl bg-
purple-500/10 border border-purple-500/20 flex
items-center justify-center text-purple-400">
<i class="fa-solid fa-volume-high text-lg"></i>
</div>
<div>
```

<h3 class="font-bold text-slate-100 text-base">3. Spatialized TTS Output</h3>
Directional Audio Routing & Prosody

Panning: Center

<label class="text-xs text-slate-400 block mb-1">Text Stream for Synthesis:</label>
<input type="text" id="tts-input-text" value="System warning: Unread security notification." class="w-full bg-slate-950 border border-slate-800 rounded-xl p-2.5 text-sm text-

```
slate-200">
</div>

<div class="grid grid-cols-3 gap-2">
<button onclick="setTTSDirection('Left')" id="dir-left" class="py-2 bg-slate-800 hover:bg-slate-700 text-slate-300 rounded-xl text-xs border border-slate-700">Left Ear</button>
<button onclick="setTTSDirection('Center')" id="dir-center" class="py-2 bg-purple-600 text-white rounded-xl text-xs font-semibold shadow">Center</button>
<button onclick="setTTSDirection('Right')" id="dir-right" class="py-2 bg-slate-800 hover:bg-slate-700 text-slate-300 rounded-xl text-xs border border-slate-700">Right Ear</button>
</div>

<div class="bg-slate-950 border border-slate-800 rounded-xl p-4 flex flex-col items-center justify-
```

```
center gap-3">
<div class="flex items-center gap-1 h-10">
<div class="w-1.5 bg-purple-500 rounded-full
wave-bar" style="animation-delay: 0.1s"></div>
<div class="w-1.5 bg-purple-500 rounded-full
wave-bar" style="animation-delay: 0.3s"></div>
<div class="w-1.5 bg-purple-500 rounded-full
wave-bar" style="animation-delay: 0.2s"></div>
<div class="w-1.5 bg-purple-500 rounded-full
wave-bar" style="animation-delay: 0.5s"></div>
</div>
<button onclick="speakSpatialTTS()" class="w-
full py-2.5 bg-purple-600 hover:bg-purple-500
text-white font-semibold rounded-xl text-xs
shadow flex items-center justify-center gap-2">
<i class="fa-solid fa-play"></i> Trigger
Spatialized Audio
</button>
</div>
</div>
```

```
</div>
</section>

<!-- 4. Refreshable Braille Display -->
<section class="glass-panel rounded-2xl p-6 flex flex-col justify-between">
<div>
<div class="flex items-center justify-between mb-4 pb-3 border-b border-slate-800">
<div class="flex items-center gap-3">
<div class="w-10 h-10 rounded-xl bg-amber-500/10 border border-amber-500/20 flex items-center justify-center text-amber-400">
<i class="fa-solid fa-braille text-lg"></i>
</div>
<div>
<h3 class="font-bold text-slate-100 text-base">4. Refreshable Braille Display</h3>
<p class="text-xs text-slate-400">8-Cell Dynamic Tactile Terminal</p>
```

```
</div>
</div>
<button onclick="refreshBrailleCells()"
class="px-3 py-1.5 rounded-lg bg-slate-800
hover:bg-slate-700 text-xs text-slate-300 border
border-slate-700">
<i class="fa-solid fa-rotate mr-1"></i> Sync
</button>
</div>

<div class="space-y-4">
<div class="bg-slate-950 border border-slate-800
rounded-xl p-4 flex flex-col items-center justify-
center gap-3">
<div id="braille-display-grid" class="grid grid-
cols-4 sm:grid-cols-8 gap-2">
<!-- Populated by JS -->
</div>
<div class="text-xs font-mono text-amber-300
mt-1" id="braille-ascii-view">ASCII: [ H E L L O !
```

```
]</div>
</div>

<div class="bg-slate-950 border border-slate-800 rounded-xl p-3 flex items-center justify-between text-xs text-slate-400">
<span>Cursor Position: <strong class="text-slate-200">Cell 01</strong></span>
<span>Status: <strong class="text-emerald-400">Synchronized</strong></span>
</div>
</div>
</div>
</section>

</div>
</div>

</main>
```

```
<!-- Master Footer -->
<footer class="border-t border-slate-800 bg-slate-900 mt-12 py-6 text-center text-xs text-slate-500">
Master Integrated Workstation & Intelligence Hub • Computer Science & Biometric Security Reference
</footer>

<!-- JavaScript Master Controller -->
<script>
// --- MASTER TAB NAVIGATION ---
function switchMasterTab(tabId) {
document.querySelectorAll('.master-module').forEach(el => el.classList.add('hidden'));
document.querySelectorAll('.master-tab').forEach(btn => {
btn.className = "master-tab px-4 py-2 rounded-xl text-xs sm:text-sm font-semibold transition-all text-slate-400 hover:text-slate-200 hover:bg-
```

```
slate-800 flex items-center gap-2 whitespace-
nowrap";
});

document.getElementById('module-' +
tabId).classList.remove('hidden');
const activeBtn =
document.getElementById('nav-' + tabId);
activeBtn.className = "master-tab px-4 py-2
rounded-xl text-xs sm:text-sm font-semibold
transition-all bg-blue-600/20 text-blue-400
border border-blue-500/40 flex items-center
gap-2 whitespace-nowrap shadow-sm";
}

// --- THEME & HIGH CONTRAST ---
let isHighContrast = false;
function toggleHighContrast() {
isHighContrast = !isHighContrast;
document.body.classList.toggle('high-contrast',
```

```
isHighContrast);
}

function toggleTheme() {
const html = document.documentElement;
const icon =
document.getElementById('themeIcon');
if (html.classList.contains('dark')) {
html.classList.remove('dark');
html.classList.add('light');
icon.textContent = '☀️';
} else {
html.classList.remove('light');
html.classList.add('dark');
icon.textContent = ' 🌙';
}
}

// --- MODULE 1 JS: FUNDAMENTALS &
GENERATIONS ---
```

```
const generationsData = {
1: { gen: "1st Generation (1940s–1950s)", tech: "Vacuum Tubes / Thermionic Valves", memory: "Magnetic Drum & Punch Cards", speed: "Milliseconds", languages: "Machine Code (Binary)", examples: "ENIAC, UNIVAC I", desc: "First generation computers relied on fragile vacuum tubes. They occupied large rooms, generated immense heat, and burned out frequently." },
2: { gen: "2nd Generation (1950s–1960s)", tech: "Bipolar Junction Transistors", memory: "Magnetic Core Memory", speed: "Microseconds", languages: "Assembly, FORTRAN, COBOL", examples: "IBM 7090, CDC 1604", desc: "Transistors replaced vacuum tubes, creating smaller, faster, and more reliable solid-state machines." },
3: { gen: "3rd Generation (1960s–1970s)", tech: "Integrated Circuits (ICs)", memory:
```

"Semiconductor RAM/ROM", speed: "Nanoseconds", languages: "BASIC, Pascal, C", examples: "IBM System/360, PDP-8", desc: "Integrated circuits combined multiple transistors onto silicon chips. Keyboards, monitors, and OS operating systems were introduced." },
4: { gen: "4th Generation (1970s–Present)", tech: "Microprocessors (VLSI/ULSI)", memory: "Semiconductor RAM, SSDs", speed: "Picoseconds / GHz", languages: "C++, Java, Python", examples: "Intel 4004/x86, Apple II, PCs", desc: "VLSI placed millions of transistors on a single CPU microchip, sparking the personal computer revolution." },
5: { gen: "5th Generation (Present & Beyond)", tech: "NPUs, AI & Quantum Qubits", memory: "NVMe Flash, Quantum Memory", speed: "TeraFLOPS / QFLOPS", languages: "AI Neural Frameworks", examples: "NVIDIA Tensor, IBM

```
Quantum", desc: "Focused on artificial
intelligence, parallel neural processing, and
quantum computing capabilities." }
};

function selectGen(num) {
const data = generationsData[num];
for (let i = 1; i <= 5; i++) {
const btn = document.getElementById(tabGen$
{i}`);
if (i === num) {
btn.className = "gen-tab px-3 py-1 rounded-lg
text-xs font-semibold bg-blue-600 text-white";
} else {
btn.className = "gen-tab px-3 py-1 rounded-lg
text-xs font-semibold text-slate-400 hover:text-
white";
}
}
```

document.getElementById('genDetailPanel').innerHTML = <div class="flex justify-between items-center mb-4 pb-3 border-b border-slate-800"> <span class="text-xs font-bold text-blue-400 uppercase">${data.gen}

Speed:

$data.speedlt; /stronggt; lt; /spangt; lt; /divgt; lt; h4class = ” text - xlfont - boldtext - whitemb - 2” gt;$

{data.tech}</h4>

<p class="text-sm text-slate-300 leading-relaxed mb-4">

$data.desclt; /pgt; lt; divclass = ” gridgrid - cols - 1sm : grid - cols - 2gap - 3text - xs” gt; lt; divclass = ” bg - slate - 950p - 3rounded - xlborderborder - slate - 800” gt; lt; stronggt; Memory : lt; /stronggt;$

{data.memory}

Languages:

$data.languageslt; /divgt; lt; divclass = ” bg - slate - 950p - 3rounded - xlborderborder - slate - 800col - span - 2” gt; lt; stronggt; Examples : lt; /stronggt;$

{data.examples}

`;

}

```
function updateSpeedDemo(val) {
document.getElementById('speedValue').innerTe
xt = (val / 10).toFixed(1) + ' GHz';
}

function runGIGODemo(isClean) {
const el =
document.getElementById('gigoResult');
if (isClean) {
el.className = "text-[11px] text-emerald-400
font-mono mt-1";
el.innerText = "Input [2 + 2] → Output: 4 (100%
Accurate)";
} else {
el.className = "text-[11px] text-rose-400 font-
mono mt-1";
el.innerText = "Input [2 + 'A'] → Output: NaN
(GIGO Error)";
}
```

```
}

// Quiz Logic
const quizQuestions = [
{ q: "Which primary characteristic means computers operate endlessly without fatigue?", options: ["Speed", "Accuracy", "Diligence", "Versatility"], correct: 2 },
{ q: "What key hardware technology defined 1st Generation computers?", options: ["Transistors", "Vacuum Tubes", "Integrated Circuits", "Microprocessors"], correct: 1 },
{ q: "What does GIGO stand for in computer accuracy?", options: ["Garbage In, Garbage Out", "Gigahertz In, Gigabytes Out", "General Input, General Output", "Great Instruction, Great Operation"], correct: 0 },
{ q: "Microprocessors with VLSI technology belong to which generation?", options: ["2nd Gen", "3rd Gen", "4th Gen", "5th Gen"], correct: 2 }
```

];
let currentQ = 0;
let quizScore = 0;

function renderQuiz() {
const q = quizQuestions[currentQ];
const container =
document.getElementById('quizContainer');
if (!q) {
container.innerHTML = <div class="text-center
py-6"><h4>Quiz Completed! Score: $
{quizScore}/4</h4><button onclick="resetQuiz()"
class="mt-3 px-4 py-2 bg-blue-600 text-white
rounded-xl text-xs font-bold">Retake</button></
div>`;
return;
}

container.innerHTML = <div class="space-y-4
text-xs sm:text-sm"> <div class="font-bold

text-white text-base">${q.q}

latex

{q.options.map((opt, idx) => `
<button onclick="answerQuiz(

{idx})" class="w-full text-left p-3 rounded-xl bg-slate-950 border border-slate-800 hover:border-blue-500 text-slate-200">${opt}</button>).join('')}

`;
}

function answerQuiz(idx) {
if (idx === quizQuestions[currentQ].correct) quizScore++;
document.getElementById('quizScoreDisplay').in

```
nerText = ``${quizScore} / 4`;
currentQ++;
renderQuiz();
}

function resetQuiz() { currentQ = 0; quizScore =
0; renderQuiz(); }


// --- MODULE 2 JS: BIOMETRIC SIMULATOR ---
function updateBioSimulation() {
const noise =
parseInt(document.getElementById('bio-
noise').value);
const light =
parseInt(document.getElementById('bio-
light').value);
const crowd =
parseInt(document.getElementById('bio-
crowd').value);
```

```
document.getElementById('bio-noise-val').innerText = noise + ' dB';
document.getElementById('bio-light-val').innerText = light > 75 ? 'Direct Glare' : 'Balanced';
document.getElementById('bio-crowd-val').innerText = crowd > 70 ? 'High' : 'Moderate';

let voice = Math.max(10, 100 - (noise * 0.8));
let vision = Math.max(20, 95 - Math.abs(light - 50) * 0.5 - (crowd * 0.2));

voice = Math.round(voice);
vision = Math.round(vision);

document.getElementById('bio-vision-score').innerText = vision + '%';
document.getElementById('bio-vision-bar').style.width = vision + '%';
```

```
document.getElementById('bio-voice-score').innerText = voice + '%';
document.getElementById('bio-voice-bar').style.width = voice + '%';
}

function resetBioSim() {
document.getElementById('bio-noise').value = 65;
document.getElementById('bio-light').value = 50;
document.getElementById('bio-crowd').value = 40;
updateBioSimulation();
}

function calculateDeployment() {
document.getElementById('calc-output').classList.remove('hidden');
}
```

```
// --- MODULE 3 JS: ACCESSIBLE WORKSTATION ---
let currentBrailleDots = [0,0,0,0,0,0];
let tactileBuffer = "";
let ttsDirection = "Center";

function startVoiceRecognition() {
const transcript = document.getElementById('voice-transcript');
transcript.innerText = "Read unread emails and summary.";
logCommand(Voice Intent: "Read unread emails and summary.", "Voice");
updateBrailleDisplayWithText("EMAILS");
}

function simulateVoiceCommand(cmd) {
document.getElementById('voice-transcript').innerText = "${cmd}";
logCommand(Voice Intent: "${cmd}", "Voice");
```

```
updateBrailleDisplayWithText(cmd);
}

function logCommand(text, src) {
const log =
document.getElementById('command-log');
const div = document.createElement('div');
div.innerHTML = <span class="text-slate-500"&gt;
[${new Date().toLocaleTimeString()}]</span>
<span class="text-blue-400">[${src}]&lt;/
span&gt; ${text}`;
log.prepend(div);
}

function toggleBrailleDot(dotNum) {
const idx = dotNum - 1;
currentBrailleDots[idx] = currentBrailleDots[idx]
=== 1 ? 0 : 1;
document.getElementById(dot-$
{dotNum}).classList.toggle('active',
```

```
currentBrailleDots[idx] === 1);
document.getElementById('chord-
display').innerText =Chord: $
{currentBrailleDots.join('')};
}

function submitBrailleChord() {
const chord = currentBrailleDots.join('');
tactileBuffer += chord === "100000" ? "a" : (chord
=== "110000" ? "b" : "x");
document.getElementById('tactile-output-
buffer').innerText = tactileBuffer;
logCommand(Braille chord [${chord}]
translated.`, "Tactile");
updateBrailleDisplayWithText(tactileBuffer);
clearBrailleChord();
}

function clearBrailleChord() {
currentBrailleDots = [0,0,0,0,0,0];
```

```
for(let i=1; i<=6; i++)
document.getElementById(dot-$
{i}).classList.remove('active');
document.getElementById('chord-
display').innerText =Chord: 000000`;
}

function setTTSDirection(dir) {
ttsDirection = dir;
document.getElementById('tts-direction-
badge').innerText = Panning:${dir};
logCommand(Audio panned to ${dir} ear., "TTS");
}

function speakSpatialTTS() {
const text = document.getElementById('tts-input-
text').value;
logCommand(Synthesizing TTS (${ttsDirection}):
"${text}", "TTS");
if ('speechSynthesis' in window) {
```

```
window.speechSynthesis.speak(new
SpeechSynthesisUtterance(text));
}
}

function refreshBrailleCells() {
updateBrailleDisplayWithText("WORKSTN");
logCommand("Refreshable braille cells
synchronized.", "Braille");
}

function updateBrailleDisplayWithText(text) {
const grid = document.getElementById('braille-
display-grid');
grid.innerHTML = '';
const upper = text.toUpperCase().padEnd(8, '
').substring(0,8);
document.getElementById('braille-ascii-
view').innerText = ASCII: [${upper.split('').join(' ')}
]`;
```

```
for(let i=0; i<8; i++) {
const char = upper[i];
let dotsHtml = '<div class="grid grid-cols-2
gap-1">';
for(let d=1; d<=6; d++) {
const active = (char.charCodeAt(0) + d) % 2 ===
0 && char !== ' ';
dotsHtml += <div class="w-2.5 h-2.5 rounded-full
${active ? 'bg-amber-400' : 'bg-slate-800'}"></
div>; } dotsHtml += '&lt;/div&gt;'; const cell =
document.createElement('div'); cell.className =
"bg-slate-900 border border-slate-800 rounded-xl
p-2 flex flex-col items-center justify-center
gap-1"; cell.innerHTML =<span class="text-[9px]
font-mono text-slate-500">C${i+1}&lt;/span&gt;$
{dotsHtml}<span class="text-xs font-bold text-
amber-300">${char}&lt;/span>;
grid.appendChild(cell);
}
```

```
}

// Initialize on load
window.onload = function() {
selectGen(1);
renderQuiz();
updateBioSimulation();
updateBrailleDisplayWithText("WORKSTN");
};
</script>
</body>
</html>
```